

Project Report

CUSTOMER BEHAVIOR ANALYSIS FOR RETAIL BUSINESS

(Using Python, MySQL Workbench, and Power BI)

1. Abstract

Customer behavior analysis plays an important role in helping retail and e-commerce companies understand purchasing trends, customer preferences, and business performance. This project focuses on analyzing customer transaction data to uncover valuable business insights that can improve revenue generation, customer retention, and marketing effectiveness.

The project involves data cleaning and exploratory data analysis using Python, storing and querying cleaned data using MySQL Workbench, and developing interactive dashboards using Power BI. Various business questions related to revenue generation, discounts, customer segmentation, subscription behavior, product performance, and demographic contribution were analyzed.

The results of this project help businesses make data-driven decisions regarding inventory management, customer targeting, discount optimization, and subscription strategies.

2. Introduction

In the modern retail industry, businesses generate huge volumes of customer transaction data daily. Analyzing this data helps organizations understand customer purchasing behavior and improve operational efficiency.

Retail companies often face challenges such as:

- Low customer retention
- Ineffective marketing campaigns
- Poor understanding of customer demand
- Weak customer segmentation
- Inefficient discount strategies

This project aims to solve these business challenges by applying data analytics techniques using Python, MySQL Workbench, and Power BI.

3. Problem Statement

The retail company wants to understand customer purchasing behavior to:

- Increase overall revenue
- Improve customer retention
- Optimize discount and promotional strategies
- Identify high-performing products and categories
- Understand customer demographics
- Improve subscription and loyalty programs

The company currently possesses raw transactional data but lacks meaningful insights that can support strategic business decisions.

4. Project Objectives

- The main objectives of this project are:
1. Perform data cleaning and preprocessing using Python.
 2. Conduct Exploratory Data Analysis (EDA).
 3. Store cleaned data in MySQL Workbench.
 4. Solve business problems using SQL queries.
 5. Create interactive dashboards in Power BI.
 6. Generate actionable business insights.

5. Tools & Technologies Used

Tool	Purpose
Python	Data Cleaning & EDA
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib & Seaborn	Visualization
MySQL Workbench	Database Storage & SQL Analysis
Power BI	Dashboard & Reporting
CSV Dataset	Data Source

6. Dataset Overview

The dataset used in this project is a structured retail/e-commerce dataset containing customer transaction information.

Dataset Characteristics

Feature Description

- TypeRetail / E-commerce
- NatureStructured (Tabular)
- LevelCustomer Transaction Level

Data Types

Numerical Columns

- Age
- Purchase Amount
- Previous Purchases
- Review Rating

Categorical Columns

- Gender
- Category
- Payment Method
- Season

- Subscription Status
- Shipping Type

7. Data Dictionary

Column Name	Description
Customer ID	Unique customer identifier
Age	Age of customer
Gender	Customer gender
Item Purchased	Product purchased
Category	Product category
Purchase Amount	Amount spent
Location	Customer location
Size	Product size
Color	Product color
Season	Season of purchase
Review Rating	Product rating
Subscription Status	Subscription membership
Shipping Type	Delivery method
Discount Applied	Whether discount applied
Previous Purchases	Past purchase count
Payment Method	Payment mode
Frequency of Purchases	Buying frequency

8. Methodology

The project followed the below workflow:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Data Export to MySQL Workbench
5. SQL Query Analysis
6. Power BI Dashboard Development
7. Business Insight Generation

9. Data Cleaning & Preprocessing using Python

The dataset contained inconsistencies, missing values, and duplicate records that required cleaning before analysis.

9.1 Data Loading

The dataset was loaded using Pandas.

```
import pandas as pd
df = pd.read_csv("customer_shopping_behavior.csv")
```

9.2 Initial Data Exploration

The following functions were used:

```
df.head()
df.info()
df.describe()
df.isnull().sum()
```

Purpose:

- Understand dataset structure
- Identify missing values
- Check data types
- Detect inconsistencies

9.3 Category Consistency Correction

Incorrect category mappings between products and categories were identified and corrected using dictionary mapping.

Purpose:

- Improve analysis accuracy
- Ensure proper product categorization

9.4 Missing Value Handling

Size Column

Different strategies were used:

- Electronics → Not Applicable
- Accessories → Free Size
- Clothing → Mode value

Review Rating

Missing ratings were filled using product-level mean ratings.

Purchase Amount

Filled using average purchase amount.

Previous Purchases

Missing values replaced with 0.

Purpose:

- Preserve data quality
- Maintain realistic business logic

9.5 Duplicate Removal

Duplicate customer records were identified and removed.

```
df.drop_duplicates(inplace=True)
```

Purpose:

- Prevent double counting
- Improve reporting accuracy

9.6 Column Standardization

```
df.columns = df.columns.str.replace(" ", "_")
```

```
df.columns = df.columns.str.lower()
```

Purpose:

- SQL-friendly naming
- Easy Power BI integration

10. Data Export to MySQL Workbench

After cleaning, the processed dataset was exported to MySQL Workbench for SQL-based analysis.

Why MySQL Workbench?

- Easy database management
- Efficient SQL querying
- Better integration for analysis workflows

Connection Process

Python libraries used:

```
pip install mysql-connector-python sqlalchemy
```

Example connection:

```
from sqlalchemy import create_engine
```

```
engine = create_engine("mysql+mysqlconnector://root:password@localhost/customer_analysis")
```

```
df.to_sql("customer_data", engine, if_exists="replace", index=False)
```

11. Exploratory Data Analysis (EDA)

EDA was performed to identify patterns, trends, and relationships in customer purchasing behavior.

Main analysis areas:

- Revenue analysis
- Customer segmentation
- Product performance
- Discount impact
- Subscription analysis
- Age-group contribution
- Shipping behavior

12. Business Questions & SQL Analysis

12.1 Which category generates the highest revenue?

Electronics generated the highest revenue (~\$486K).

Business Insight

Electronics products contribute significantly to company revenue and should receive focused inventory and marketing investment.

12.2 Are discounts increasing purchase value?

Customers using discounts spent more on average (\$219) compared to non-discount purchases (\$182).

Business Insight

Discount campaigns positively influence customer spending behavior.

12.3 Revenue by Gender

- Male: ~\$391K
- Female: ~\$352K
- Other: ~\$251K

Business Insight

Male customers contributed the highest overall revenue.

12.4 Top Rated Products

Highest-rated products:

- Gloves
- Sandals
- Boots

Lowest-rated products:

- Phone
- Headphones
- Watch

Business Insight

High-rated products can be promoted more aggressively, while low-rated products may require quality improvement.

12.5 Subscription Analysis

Subscribed customers spend significantly more than non-subscribers.

Customer Type Avg Spend

Subscribers \$270

Non-Subscribers \$166

Business Insight

Subscription programs improve customer lifetime value.

12.6 Customer Segmentation

Customers were divided into:

- Loyal Customers
- Returning Customers
- New Customers

Results:

- Loyal Customers: 3085
- Returning Customers: 1354
- New Customers: 561

Business Insight

Loyal customers form the major revenue-driving segment.

12.7 Revenue Contribution by Age Group

Age Group Revenue

51+	Highest
36–50	Second Highest
26–35	Moderate
18–25	Lowest

Business Insight

Older customers contribute the highest revenue.

13. Power BI Dashboard

Interactive dashboards were created to visualize business insights.

Dashboard Features

- KPI Cards
- Revenue Analysis
- Category Analysis
- Customer Segmentation
- Shipping Insights
- Seasonal Analysis
- Payment Method Analysis
- Product Ratings

Key KPIs

KPI	Value
Total Customers	5000
Unique Items	30

KPI	Value
Total Spend	994K
Average Spend	198
Average Rating	3.62

14. Key Findings

1. Electronics is the highest revenue-generating category.
2. Discounts increase average purchase value.
3. Subscribers spend significantly more than non-subscribers.
4. Loyal customers contribute major repeat revenue.
5. Customers aged 51+ generate the highest revenue.
6. Standard shipping is the most frequently used shipping type.
7. Debit cards are the most preferred payment method.

15. Business Recommendations

Marketing Recommendations

- Increase targeted discounts for high-spending customers.
- Promote subscription programs aggressively.

Product Recommendations

- Invest more in electronics inventory.
- Improve quality of low-rated products.

Customer Retention Recommendations

- Reward loyal customers with exclusive benefits.
- Create personalized campaigns for returning customers.

Sales Strategy Recommendations

- Target older age groups with premium offerings.
- Optimize seasonal campaigns based on demand trends.

16. Conclusion

This project successfully analyzed customer purchasing behavior using Python, MySQL Workbench, and Power BI.

The analysis provided valuable insights into:

- Customer spending habits
- Product performance
- Discount effectiveness
- Customer loyalty
- Subscription behavior

The project demonstrates how data analytics can help retail businesses make smarter decisions, improve customer retention, optimize marketing strategies, and increase profitability.

17. Future Scope

Future improvements may include:

- Machine Learning-based customer prediction
- Recommendation systems
- Sales forecasting
- Real-time dashboard integration
- Customer churn prediction

18. References

- Python Documentation
- Pandas Documentation
- MySQL Workbench Documentation
- Microsoft Power BI Documentation

